

Lecture: Ethics

Actuarial Data Science - Open Learning Resource
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- Topic 2: How to evaluate fairness and mitigate indirect discrimination and bias in the context of AI and Big Data?



Recommended Reading

- [The Discriminating \(Pricing\) Actuary \(Frees and Huang 2023\)](#)
- [Antidiscrimination Insurance Pricing: Regulations, Fairness Criteria, and Models \(Xin and Huang 2024\)](#)



Learning Objectives

Ethical considerations are central to modern actuarial and data science practice, especially in the context of AI and automated decision-making. In this lecture, we explore key ethical principles, their implications for insurance pricing, and how to balance fairness, regulation, and business objectives.

- Understand the key principles of AI ethics, including fairness, transparency, accountability, and privacy.
- Explain the role of discrimination and fairness in insurance, and distinguish between social and economic perspectives.
- Evaluate different approaches to mitigating bias and discrimination, including input-based and output-based methods.
- Describe common quantitative fairness criteria and discuss their applicability in actuarial contexts.
- Assess trade-offs between fairness, accuracy, and business objectives in real-world applications.



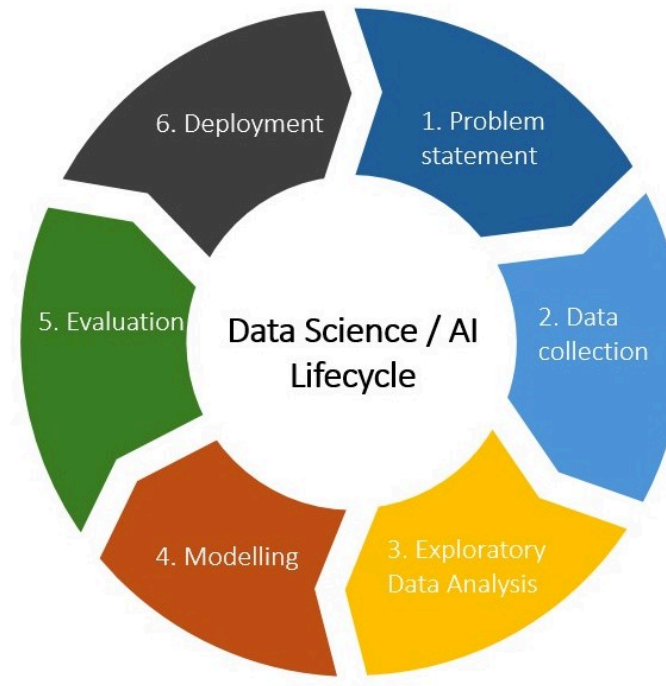
Revisiting the Data Science Lifecycle

Q: What is the most frequent failure in data analysis?

A: Problem formulation

“The most frequent failure in data analysis is mistaking the type of question being considered.”

— Leek and Peng (2015)



Data Science (AI) Lifecycle

Major Principles of AI Ethics

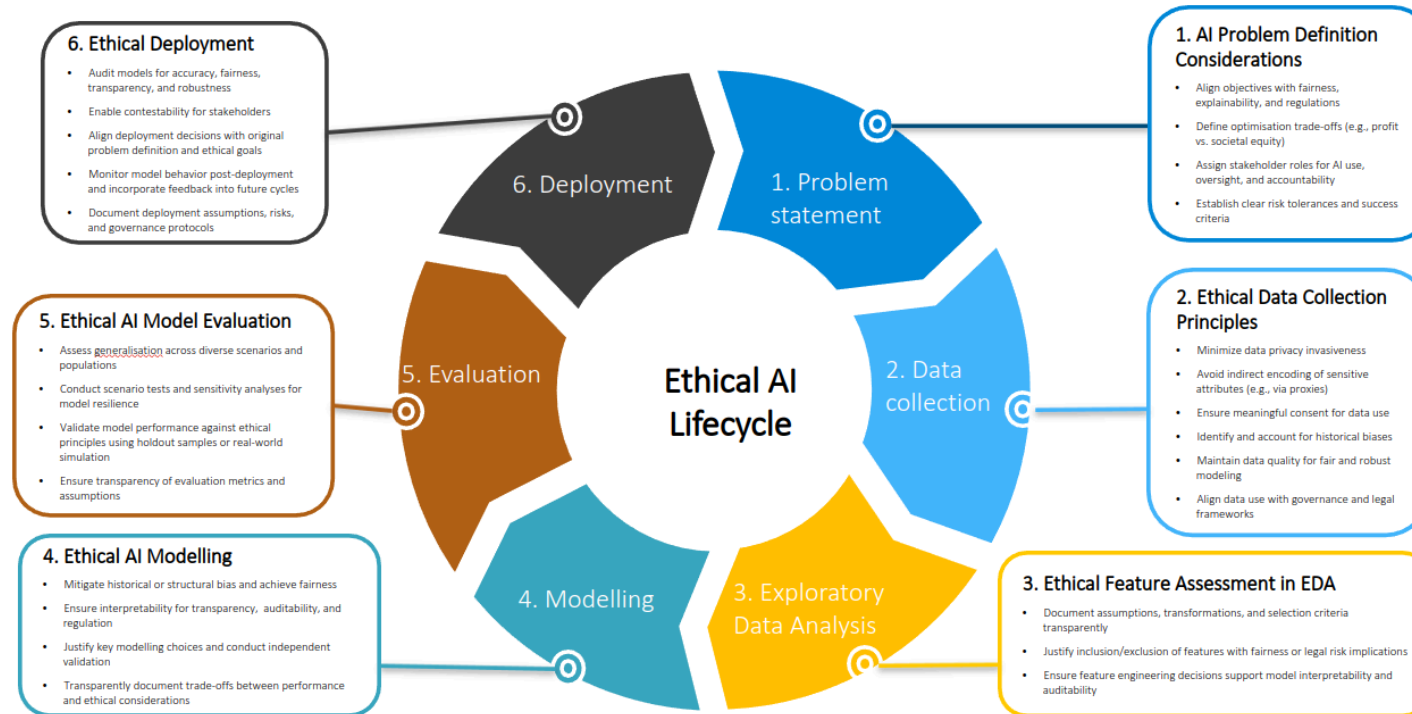
- **Fairness and non-discrimination:** Mitigate both direct and indirect discrimination embedded in data or model design.
- **Transparency and explainability:** Ensure decisions are interpretable not only to developers, but also to customers and regulators.
- **Accountability:** Clarify who is responsible when AI systems make errors (e.g. actuaries, developers, or firms).
- **Privacy and data ethics:** Address issues related to third-party data, telematics, and social media sources.
- **Contestability:** Provide individuals with meaningful channels to challenge and appeal automated decisions
- **Stability and robustness:** Ensure AI models remain reliable and perform as expected under real-world conditions.



Ethical AI Lifecycle

“The actuarial profession should lead, not follow, in building public trust in AI systems.”

— The Actuary, IFoA



Ethical AI Lifecycle.



Regulatory Responses to AI and Discrimination in Insurance



Colorado SB21-169.

Colorado SB21-169: Protecting Consumers from Unfair Discrimination in Insurance Practices

Proposed Insurance Circular Letter

January 17, 2024

TO: All Insurers Authorized to Write Insurance in New York State, Licensed Fraternal Benefit Societies, and the New York State Insurance Fund

RE: Use of Artificial Intelligence Systems and External Consumer Data and Information Sources in Insurance Underwriting and Pricing

STATUTORY AND REGULATORY REFERENCES: N.Y. Ins. Law §§ 308, 309, 1501, 1503, 1604, 1702, 1717, 2303, 3221, 3425, 3426, 4224, and 4305, and Articles 24 and 26; 11 NYCRR 82; 11 NYCRR 89; 11 NYCRR 90; 11 NYCRR 243

I. Purpose and Background

1. The New York State Department of Financial Services ("Department") is committed to innovation and the responsible use of technology to improve financial access and contribute to the safety and stability of insurance markets. The Department expects that insurers use of emerging technologies such as artificial intelligence will be conducted in a manner that complies with all applicable federal and state laws, rules, and regulations.
2. The use of external consumer data and information sources ("ECDIS") and artificial intelligence systems ("AIS") can both benefit insurers and consumers alike by simplifying and expediting insurance underwriting and pricing processes, and potentially result in more accurate underwriting and pricing of insurance. At the same time, ECDIS may reflect systemic biases and its use can reinforce and exacerbate inequality. This raises significant concerns about the potential for unfair adverse effects or discriminatory decision-making. ECDIS may also have variable accuracy and reliability and may come from entities that are not subject to regulatory oversight and consumer protections. Furthermore, the self-learning behavior of AIS increases the risks of inaccurate, arbitrary, capricious, or unfairly discriminatory outcomes that may disproportionately affect vulnerable communities and individuals or otherwise undermine the insurance marketplace in New York.
3. Therefore, it is critical that insurers who utilize such technologies establish a proper governance and risk management framework to mitigate the potential harm to consumers and comply with all relevant legal obligations. The purpose of this circular letter ("Circular Letter") is to identify DFS's expectations that all insurers authorized

New York DFS Circular Letter.

New York DFS – Use of Artificial Intelligence Systems and External Consumer Data and Information Sources in Insurance Underwriting and Pricing



Guidance on AI and Discrimination in Insurance (Australia)



Australian Human Rights Commission (AHRC)

Guidance Resource: AI and Discrimination in Insurance (AHRC)



From Bias to Fairness

How can bias related to protected attributes lead to unfair discrimination — and how can it be mitigated?



Topic 1: What are the principles to assess the appropriateness of insurance discrimination (differentiation)?

Adapted from Frees and Huang (2023), *North American Actuarial Journal*

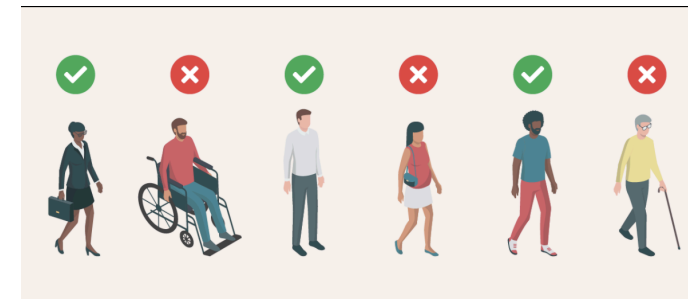


Background

- Discrimination is widespread in society.
- Insurance pricing is interesting because the entire industry is based on discrimination.
- However, certain characteristics (e.g., ethnicity, religion, heritage) are typically prohibited.

Where does discrimination arise in insurance?

- Issuance, renewal, or cancellation
- Coverage
- Underwriting and pricing
- Marketing
- Claims processing



Discrimination is widespread (Source: [CBIA](#)).

Fairness Depends on Context

Is insurance a **social good** or an **economic commodity**?

Is risk pooling based on **subsidising solidarity** or **chance solidarity**?



Insurance as a Social Good

- Benefits the general public (e.g., clean air, clean water)
- Non-excludable
- Mandatory insurance shifts insurance toward a social good (social safety net)



Examples

- Health insurance is often viewed as a social good
- Compulsory third-party auto insurance is a social good.
- Catastrophe insurance is treated as a social good by some jurisdiction.



Insurance as an Economic Commodity

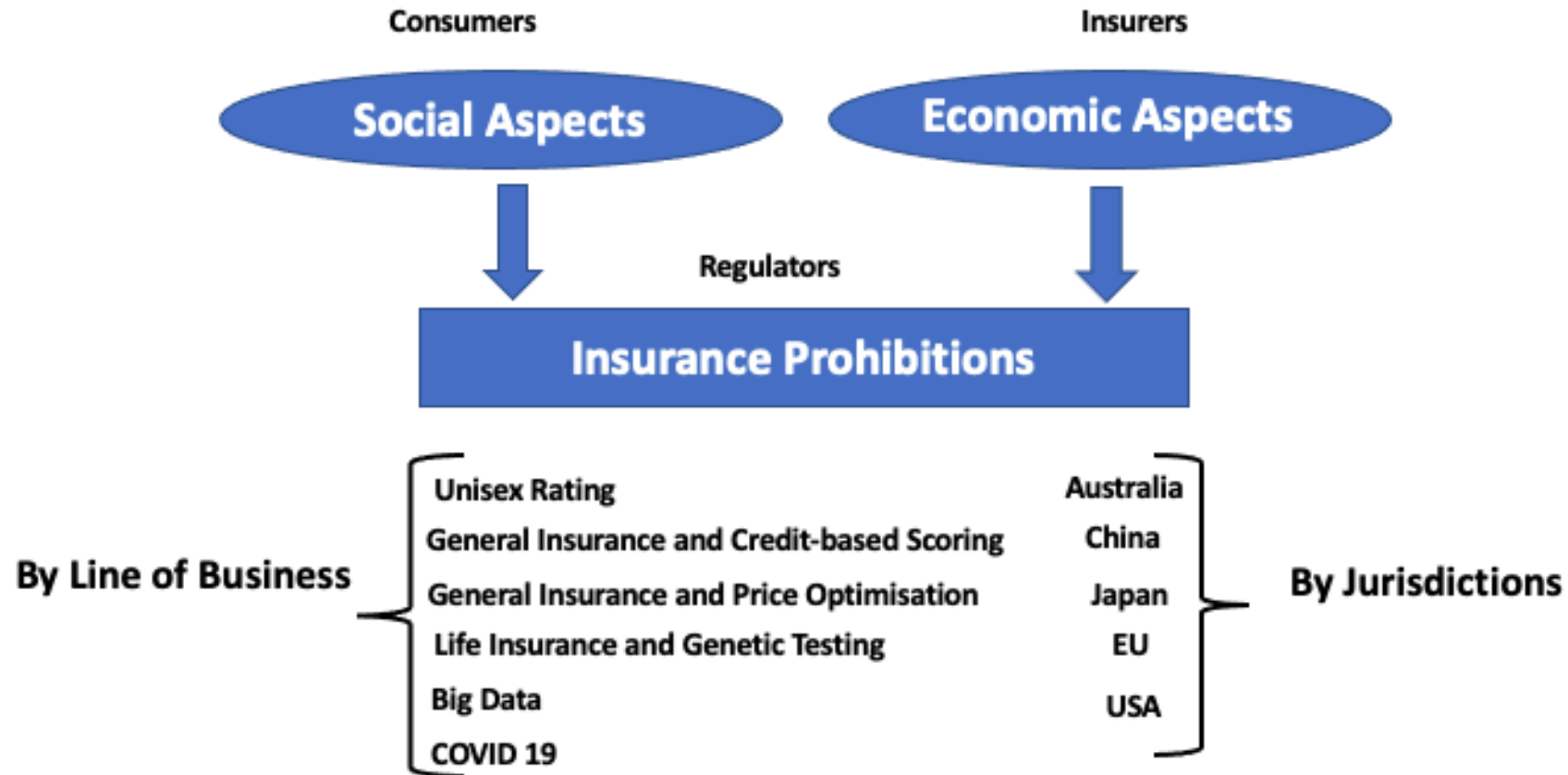
- Voluntary insurance shifts insurance toward an economic commodity.
- Market price depends on the forces of supply and demand
- “Free” market (limited regulation)
- Risk classification is advocated to address adverse selection, moral hazard, and improve economic efficiency



Examples

- Life insurance is often viewed as a private (non-public) product, an economic commodity.
- Voluntary auto insurance is an economic commodity.
- Long-term care and disability insurance fall in between.

Balancing Social Justice and Economic Efficiency



How to balance the social justice and economic efficiency?



Is it fair to use that rating factor?

- **Control:** e.g., ownership of a sports car; many sensitive attributes are uncontrollable (gender, race, ethnicity, nationality, etc.)
- **Mutability:** Does the variable change over time (such as age) or stay fixed?
- **Causality:** It is generally acceptable to use a variable if it is known to cause an insured event (e.g., cancer in life insurance).
- **Statistical Discrimination:** A variable must have some predictive value of the underlying risk. A necessary, but not sufficient, condition.
- **Limiting or Reversing the Effects of Past Discrimination.** e.g., discriminating based on skin color is more problematic than based on eye color.
- **Inhibiting Socially Valuable Behavior:** e.g., whether to participate in genetic testing research when insurers make decisions based on genetic test results.



Indirect Discrimination: A Grey Area in Regulation

- Direct discrimination is prohibited, but indirect discrimination using proxies or more complex and opaque algorithms is not clearly regulated or assessed.
- Do we have a clear definition of indirect discrimination?
- How do we assess and mitigate indirect discrimination?



How to Mitigate Discrimination and Achieve Fairness?

- Input-based approach:
 - Restricting the use of protected or proxy variables (*fairness through unawareness*)
 - Specifying the feature importance for pricing (California regulations)
- Output-based approach:
 - Define quantitative fairness criteria
 - Protected attributes are not allowed (no direct discrimination)
 - Protected attributes are allowed as rating factors



Topic 2: How to evaluate fairness and mitigate indirect discrimination and bias in the context of AI and Big Data?

Adapted from Xin and Huang (2024), *North American Actuarial Journal* (Best Paper Award)



Existing Regulations

- No regulation
- Restriction on the use of a protected variable
- Prohibition on the use of a protected variable
- Restriction on the use of a proxy variable
- Prohibition on the use of a proxy variable
- Disparate impact standard
- Community rating
- Affirmative action



Quantitative Fairness Criteria

- **Association-based**

- Individual fairness
- Group fairness
 - Independence
 - Separation
 - Sufficiency (calibration)

- **Causality-based**

- Counterfactual fairness



Questions to consider

- Are they suitable for the insurance context?
- Are they compatible with each other?
- Which one should be used?

Further reading: Lindholm et al. (2022); Baumann and Loi (2023); Araiza Iturria, Hardy, and Marriott (2024); Charpentier (2024); Lindholm et al. (2024); Xin and Huang (2024); Côté, Côté, and Charpentier (2025); **CAS Research Paper Series**; SOA Research Paper Series



Fairness Criteria for Insurance Pricing

- Let X_P denote the protected attribute, which is a categorical variable and has only two groups $X_P = \{a, b\}$.
- Let X_{NP} denote other available (non-protected) attributes, and hence the feature space is $X = \{X_P, X_{NP}\}$.
- Let $\hat{Y}$ denote the predictor or the decision outcome of interest, $\hat{Y} \in \mathbb{R}$. In our context, $\hat{Y}$ is the premium charged by the insurer, and in this paper, we assume that $\hat{Y}$ is approximately equal to the pure premium and ignore any expenses or profit loadings.
- Let Y denote the observed outcome of interest, $Y \in \mathbb{R}$. Note that Y is not known when the policy is issued, Y is a measure of real claim experience observed by the insurer over a given period after policy issuance.



Fairness Criteria – Individual Fairness

Definition 1. Fairness through Unawareness (FTU): Fairness is achieved if the protected attribute X_P is not explicitly used in calculating the insurance premium $\hat{Y}$.

Definition 2. Fairness through Awareness (FTA): A predictor $\hat{Y}$ satisfies fairness through awareness if it gives similar predictions to similar individuals (Dwork et al. 2012; Kusner et al. 2017).

Definition 3. Counterfactual Fairness (CF): A predictor $\hat{Y}$ is counterfactually fair for an individual if “its prediction in the real world is the same as that in the counterfactual world where the individual had belonged to a different demographic group” (Kusner et al. 2017; Wu, Zhang, and Wu 2019) or, mathematically, given $X = x$ and $X_P = a$, for all y and for simplicity, X_P has only two groups $\{a, b\}$, a predictor $\hat{Y}$ is counterfactually fair if

$$\mathbb{P}(\hat{Y}_{X_P \leftarrow b}(U) = y \mid X_{NP} = x, X_P = b) = \mathbb{P}(\hat{Y}_{X_P \leftarrow a}(U) = y \mid X_{NP} = x, X_P = b).$$



Fairness Criteria – Individual Fairness (continued)

Definition 4. Controlling for the Protected Variable (CPV): As defined in definition 6 in Lindholm et al. (2022), a *discrimination-free price* for Y with respect to X_{NP} is defined by

$$h^*(X_{NP}) := \int_{x_P} \mathbb{E}[Y \mid X_{NP}, x_P] d\mathbb{P}^*(x_P),$$

where $\mathbb{P}^*(x_P)$ is defined on the same range as $\mathbb{P}(x_P)$.



Fairness Criteria – Group Fairness

Definition 5. Demographic Parity (DP): A predictor $\hat{Y}$ satisfies demographic parity if

$$\mathbb{P}(\hat{Y} \mid X_P = a) = \mathbb{P}(\hat{Y} \mid X_P = b).$$

Definition 6. Relaxed Demographic Parity (RDP): A predictor $\hat{Y}$ satisfies relaxed demographic parity or has no disparate impact if the following ratio is above certain threshold τ (Feldman et al. 2015):

$$\frac{\mathbb{P}(\hat{Y} = \hat{y} \mid X_P = b)}{\mathbb{P}(\hat{Y} = \hat{y} \mid X_P = a)} > \tau.$$



Fairness Criteria – Group Fairness (continued)

Definition 7. Conditional Demographic Parity (CDP): A predictor $\hat{Y}$ satisfies conditional demographic parity if

$$\mathbb{P}(\hat{Y} \mid X_{NP_{\text{legit}}} = x_{NP_{\text{legit}}}, X_P = a) = \mathbb{P}(\hat{Y} \mid X_{NP_{\text{legit}}} = x_{NP_{\text{legit}}}, X_P = b),$$

where $X_{NP_{\text{legit}}}$ denotes a subset of “legitimate” attributes within unprotected attributes in the feature space ($X_{NP_{\text{legit}}} \subseteq X_{NP} \subset X$) that are permitted to affect the outcome of interest ([Corbett-Davies et al. 2017](#); [Verma and Rubin 2018](#)).



Regulations, Fairness Criteria, and Models

TABLE 1
Comparison between Different Regulations

Regulation	Fairness criteria	Representative model
No regulation	Neither	M0
Restriction on a protected variable	Neither	M0*
Prohibition on a protected variable	Individual	MU or MC
Restriction on a proxy variable	Individual	MU*
Prohibition on a proxy variable	Individual	MU*
Disparate impact standard	Group	MDP or MCDP
Community rating	Group	MDP or MCDP
Affirmative action	Neither	None

Comparison between different regulations.



Questions to Tackle



Which fairness notion to use?

- Proxy discrimination (omitted variable bias)
- Conditional demographic parity
- Equalized odds (error rate parity)
- Well-calibration



How to achieve it?

- Fairness in cost prediction or market pricing?
- Fairness in terms of prices or markups?
- What if protected variables are unobserved?



What is the impact of fairness on stakeholders?

- Consumer welfare vs. firm profit
- Welfare across different groups (e.g., males vs. females)



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